Hands-on session: day 7

MD simulations with LAMMPS

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Goals of this session

Understand how to

➔ run MD simulations with LAMMPS

➔ equilibrate the system

➔ perform production run

➔ play with the role of thermostats

Turn on the VM!

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➔ Introduction to the water models

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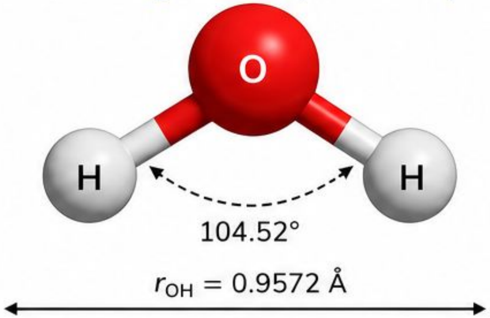
TIP3P

3-site rigid model (constrained)



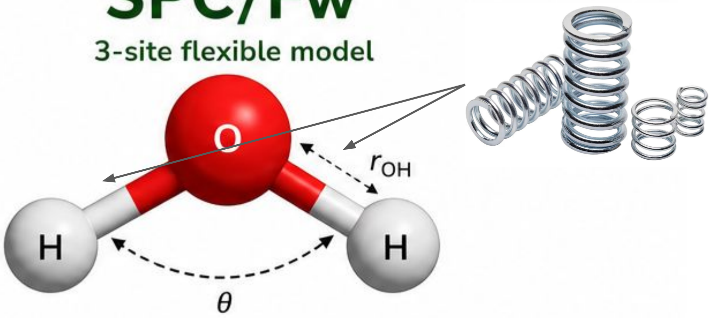
Parameter	Symbol	TIP3P value
Number of sites	—	3 (rigid)
Charge on O (oxygen)	q_O	−0.834 e
Charge on H (hydrogen)	q_H	+0.417 e
O–H bond length	r_{OH}	0.9572 Å
Equilibrium O–H bond length	r^0_{OH}	0.9572 Å
H–O–H bond angle	θ	104.52°
Equilibrium H–O–H angle	θ_0	104.52°

Constrained with SHAKE (rigid water)

In simulations, O–H bonds and H–O–H angle are constrained to their equilibrium values using the SHAKE algorithm.

SPC/Fw

3-site flexible model



Parameter	Symbol	SPC/Fw value
Number of sites	—	3 (flexible)
Charge on O (oxygen)	q_O	−0.82 e
Charge on H (hydrogen)	q_H	+0.41 e
O–H bond length	r_{OH}	Variable (flexible)
Equilibrium O–H bond length	r^0_{OH}	1.0000 Å
H–O–H bond angle	θ	Variable (flexible)
Equilibrium H–O–H angle	θ_0	109.47°

Flexible (no constraints)

In simulations, O–H bonds and H–O–H angle are allowed to fluctuate according to the force field.
No SHAKE constraints are applied.

When to use …

TIP3P

- ➔ Easier to compare with standard force-field benchmarks
- ➔ Allows larger time steps due to constrained fast vibrations
- ➔ Computationally simpler and more robust
- ➔ Good choice for electrostatics and long-range solver validation

SPC/Fw

- ➔ Useful when intramolecular flexibility may affect the dynamics
- ➔ Better suited for studying:
  - ◆ vibrational effects
  - ◆ coupling between intra- and intermolecular motion
  - ◆ dynamical properties such as VACF and diffusion
- ➔ More demanding numerically because of fast bonded vibrations

References

The Journal of
Chemical Physics

RESEARCH ARTICLE | JANUARY 10 2006

Flexible simple point-charge water model with improved liquid-state properties

Yujie Wu; Harald L. Tepper; Gregory A. Voth

Check for updates

J. Chem. Phys. 124, 024503 (2006)
https://doi.org/10.1063/1.2136877

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Articles You May Be Interested In

An improved Polarflex water model

[Link paper]

How to set it up in LAMMPS?

Version: **release**

git info: 30Mar2026

Search docs

USER GUIDE

1. Introduction
2. Install LAMMPS
3. Build LAMMPS
4. Run LAMMPS
5. Errors
6. Commands
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8. Optional packages
9. Auxiliary tools

10. Howto discussions

10.1. General howto

10.4.5. TIP3P water model

The TIP3P water model as implemented in CHARMM (MacKerell) specifies a 3-site rigid water molecule with charges and Lennard-Jones parameters assigned to each of the three atoms.

One suitable pair style with cutoff Coulomb would for instance be:

- pair_style lj/cut/coul/cut

These commands are examples for a long-range Coulomb model:

- pair_style lj/cut/coul/long
- pair_style lj/cut/coul/long/soft
- kspace_style pppm
- pair_style lj/long/coul/long
- kspace_style pppm/disp

10.4.8. SPC and SPC/E water model

The SPC water model specifies a 3-site rigid water molecule with charges and Lennard-Jones parameters assigned to each of the three atoms. In LAMMPS the fix shake command can be used to hold the two O-H bonds and the H-O-H angle rigid. A bond style of *harmonic* and an angle style of *harmonic* or *charmm* should also be used.

One suitable pair style with cutoff Coulomb would for instance be:

- pair_style lj/cut/coul/cut

These commands are examples for a long-range Coulomb model:

- pair_style lj/cut/coul/long
- pair_style lj/cut/coul/long/soft
- kspace_style pppm
- pair_style lj/long/coul/long

Link TIP3P

Link SPC

How to set it up in LAMMPS?

tip3p_water.mol

```
LAMMPS molecule file for flexible TIP3P water

3 atoms
2 bonds
1 angles

Coords

1 0.000000 0.000000 0.000000
2 0.957200 0.000000 0.000000
3 -0.239987 0.926627 0.000000

Types

1 1
2 2
3 2

Charges

1 -0.834
2 0.417
3 0.417

Bonds

1 1 1 2
2 1 1 3

Angles

1 1 2 1 3
```

tip3p_ewald_params.lmp

```
# TIP3P force-field parameters — rigid model via SHAKE constraints.
# SHAKE uses r0 and theta0 as constraint targets; spring constants are not exercised.
# Ref: Jorgensen et al., J. Chem. Phys. 79, 926 (1983).
# Units: real  |  atom types: 1=O, 2=H

# Partial charges [|e|] — molecule is neutral: -0.834 + 2*0.417 = 0
set type 1 charge -0.834   # O
set type 2 charge  0.417   # H

# LJ + long-range electrostatics (Ewald), 9 Å real-space cutoff
pair_style lj/cut/coul/long 9.0
kspace_style ewald 1.0e-6
pair_modify tail yes        # analytic LJ tail correction

# LJ: pair_coeff I J epsilon[kcal/mol] sigma[Å]
# LJ only on oxygen; hydrogen sites carry charge only
pair_coeff 1 1 0.1020 3.188   # O-O
pair_coeff 1 2 0.0    1.0     # O-H  (no LJ)
pair_coeff 2 2 0.0    1.0     # H-H  (no LJ)
```

How to set it up in LAMMPS?

spcfw_water.mol

```
LAMMPS molecule file for SPC/Fw water

3 atoms
2 bonds
1 angles

Coords

1 0.000000 0.000000 0.000000
2 1.012000 0.000000 0.000000
3 -0.399318 0.929886 0.000000

Types

1 1
2 2
3 2

Charges

1 -0.82
2 0.41
3 0.41

Bonds

1 1 1 2
2 1 1 3

Angles

1 1 2 1 3
```

spcfw_params.lmp

```
# SPC/Fw force-field parameters — flexible water model.
# Bonds and angles are integrated via the equations of motion (no SHAKE).
# Charges are set in the molecule file (spcfw_water.mol): O=-0.82, H=+0.41 |e|.
# Ref: Wu, Tepper, Voth, J. Chem. Phys. 124, 024503 (2006).
# Units: real  |  atom types: 1=O, 2=H

# LJ + long-range electrostatics (Ewald), 9 Å real-space cutoff
pair_style lj/cut/coul/long 9.0
kspace_style ewald 1.0e-6
pair_modify tail yes        # analytic LJ tail correction

# LJ: pair_coeff I J epsilon[kcal/mol] sigma[Å]
# LJ only on oxygen; hydrogen sites carry charge only
pair_coeff 1 1 0.1554253 3.165492   # O-O
pair_coeff 1 2 0.0       1.0        # O-H  (no LJ)
pair_coeff 2 2 0.0       1.0        # H-H  (no LJ)

# Bonds — harmonic: U = k*(r-r0)^2
# bond_coeff: k[kcal/mol/Å²]  r0[Å]
bond_style harmonic
bond_coeff 1 529.581 1.012      # O-H

# Angles — harmonic: U = k*(theta-theta0)^2
# angle_coeff: k[kcal/mol/rad²]  theta0[deg]
angle_style harmonic
angle_coeff 1 37.95 113.24      # H-O-H
```

How can we reproduce the results for these models?

Two steps:

1) **Equilibration** - guide the system to an equilibrium configuration
2) **Production** - producing data we can use to compute observables
    a) NVE - study what happens with conserved energy
    b) NVT - study the effect of thermostats

Note: the paper does it in NPT - some values are going to be a bit different

Practical note: activate the metatomic environment with **workon metatomic**

Equilibration - Exercise 1 (~30 min)

NVT: Nose-Hoover thermostat - notebook on how to perform equilibration

Folder: day-07-md-simulations/exercise_1

Some questions (in the notebook as well):

➔ **Thermostat speed**: which of your three *tdamp* values equilibrates fastest? Does the stiffest thermostat show visible high-frequency oscillations in T(t)?

➔ **SPC/Fw vs TIP3P**: do the two models equilibrate at the same rate for the same *tdamp*? If not, what physical difference might explain it? (Hint: think about the number of degrees of freedom and the effect of SHAKE.)

➔ **Energy stability**: does E_tot settle at the same absolute value for all three *tdamp* settings within the same model? Should it? Does the absolute value differ between SPC/Fw and TIP3P, and why? (Hint for the second part: look at the potential for the models)

➔ **Convergence and practical choice:** In the longer run at *tdamp* = 1000 fs, does each model appear converged, or is there still a visible drift in T(t) or E_tot(t) by the end of the trajectory?

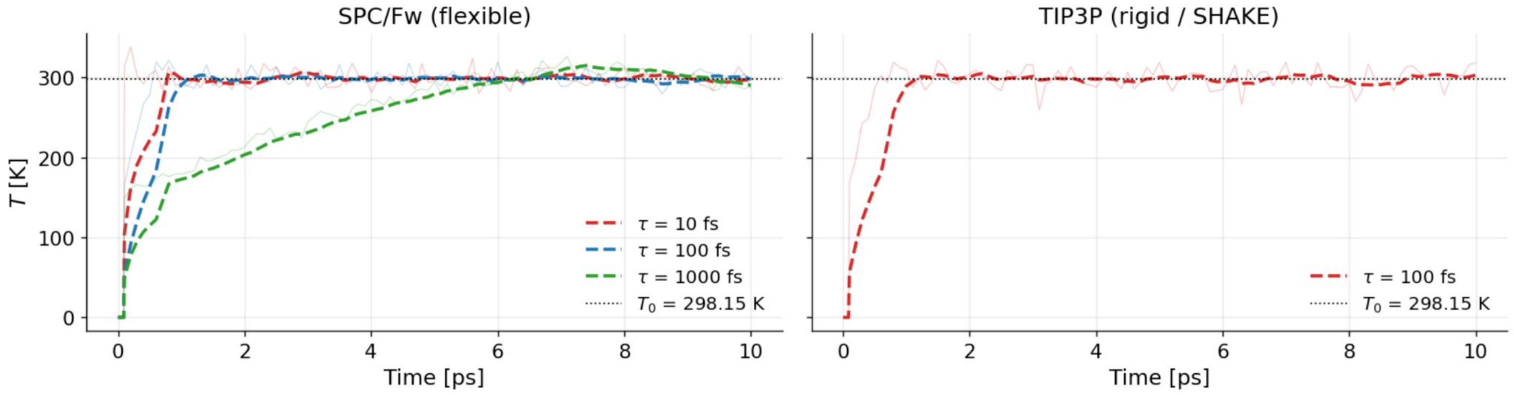
Combining Part A with the longer *tdamp* = 1000 fs run, what value of *tdamp* would you say is sufficient to reach convergence without making equilibration unnecessarily slow? Justify your answer.

Equilibration - Discussion exercise 1

➔ **Thermostat speed**: which of your three *tdamp* values equilibrates fastest? Does the stiffest thermostat show visible high-frequency oscillations in T(t)?

$$Q \propto \tau^2$$

increasing tdamp (tau), increase the inertia of the thermostat and decrease its strenght



Equilibration - Discussion exercise 1

➔ **SPC/Fw vs TIP3P**: do the two models equilibrate at the same rate for the same *tdamp*? If not, what physical difference might explain it? (Hint: think about the number of degrees of freedom and the effect of SHAKE.)

Number of active degrees of freedom:

➔ **SPC/Fw** $N_f = 3 \times 216 \times 3 - 3 = 1941$

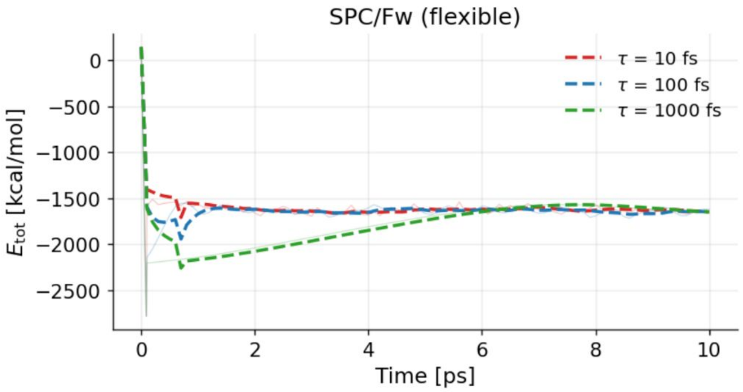
➔ **TIP3P** $N_f = 216 \times 6 - 6 = 1290$

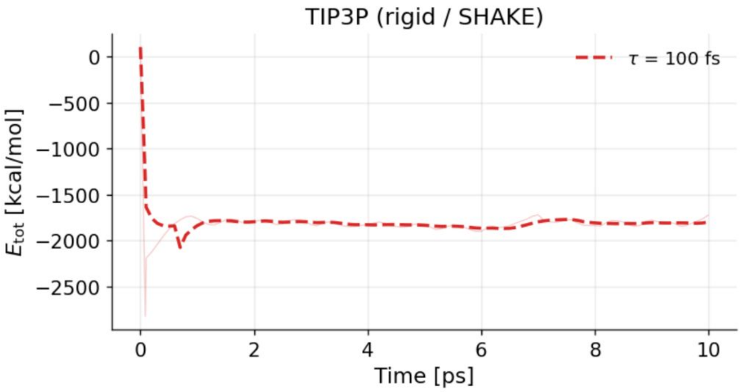
$Q = N_f k_B T_0 \tau^2$ *thermostat inertia is not the same even for identical tdamp*

Equilibration - Discussion exercise 1

➔ **Energy stability**: does E_tot settle at the same absolute value for all three *tdamp* settings within the same model? Should it? *Yes it's the same system.*

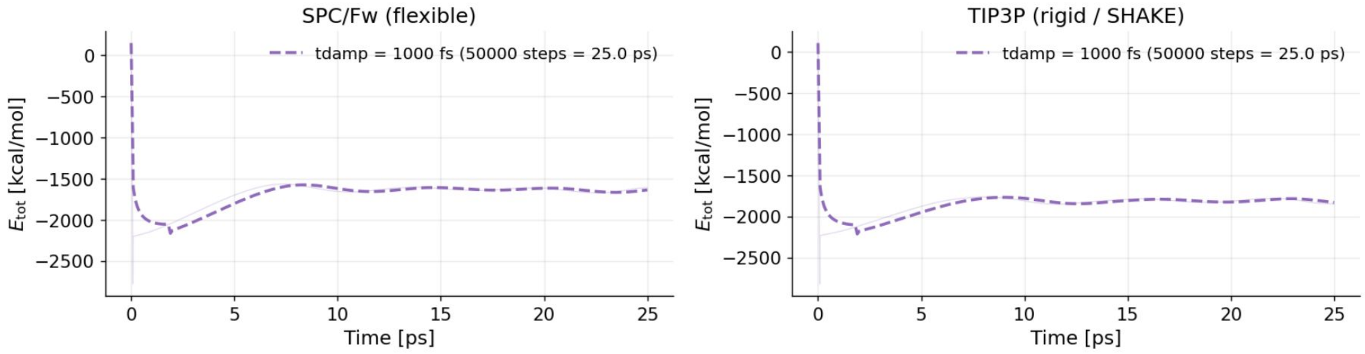
➔ Does the absolute value differ between SPC/Fw and TIP3P, and why? *Different value because different energy expressions for the two models*





Equilibration - Discussion exercise 1

➔ **Convergence and practical choice:** In the longer run at *tdamp* = 1000 fs, does each model appear converged, or is there still a visible drift in T(t) or E_tot(t) by the end of the trajectory? Combining Part A with the longer *tdamp* = 1000 fs run, what value of *tdamp* would you say is sufficient to reach convergence without making equilibration unnecessarily slow?



Production NVE - Exercise 2 (~15 min)

NVE - computing diffusion coefficient via vacf or msd

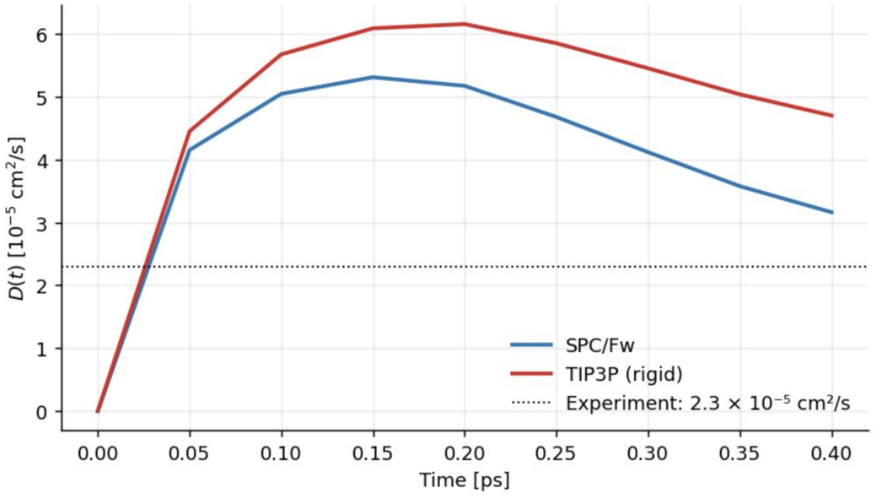
Folder: day-07-md-simulations/exercise_2

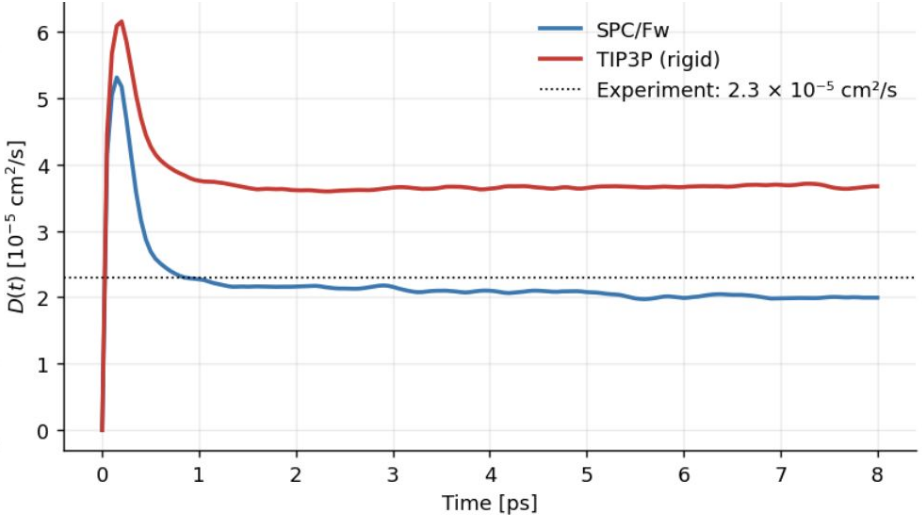
Some questions (in the notebook as well):

- ➔ **Window sensitivity (VACF):** what happens to your estimate of D_VACF if you cut the integral too early (before the plateau) or too late (deep into the noise)? Quantify the effect by trying two additional values of `VACF_TMAX_FS`.
- ➔ **Window sensitivity (MSD):** move `MSD_FIT_START_FS` into the ballistic regime (e.g. 100 fs). How much does D_MSD change? What does this tell you about the importance of the fit window choice?
- ➔ **Consistency:** Do D_VACF and D_MSD agree for the same water model? If not, which window is more likely to be wrong, and why?
- ➔ **Model comparison.** Which model reproduces the experimental D = 2.30 x 10^-5 cm²/s better? Is this result surprising given that TIP3P was parameterised to reproduce structural properties, not dynamical ones?

Production NVE - Discussion exercise 2

➔ **Window sensitivity (VACF):** what happens to your estimate of D_VACF if you cut the integral too early (before the plateau) or too late (deep into the noise)? Quantify the effect by trying two additional values of VACF_TMAX_FS.

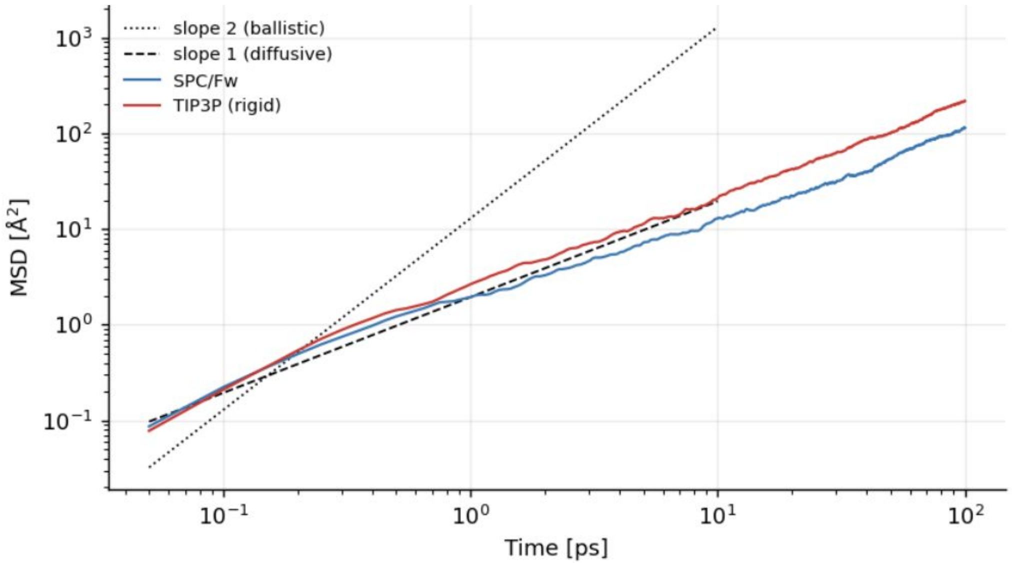




Production NVE - Discussion exercise 2

➔ **Window sensitivity (MSD):** move `MSD_FIT_START_FS` into the ballistic regime (e.g. 100 fs). How much does D_MSD change? What does this tell you about the importance of the fit window choice?

This can overestimate D. The correct starting point is after the ballistic-to-diffusive crossover (~1–2 ps for water).



Production NVE - Discussion exercise 2

➔ **Consistency:** Do D_VACF and D_MSD agree for the same water model? If not, which window is more likely to be wrong, and why?

D_VACF and D_MSD should agree within ~5–10% when both windows are correct. If they disagree, MSD window to be checked. the VACF plateau easy to spot.

	D_VACF [1e-5 cm²/s]	D_MSD [1e-5 cm²/s]	D_exp [1e-5 cm²/s]
model			
SPC/Fw	2.198	2.005	2.3
TIP3P (rigid)	3.730	3.751	2.3

```
VACF_TMAX_FS = {
    "spcfw": 1200,    # e.g. 3000
    "tip3p": 1200,    # e.g. 2000
}

# MSD linear fit window [fs]
# Choose the range where MSD is alr
# but still well-averaged (not too
MSD_FIT_START_FS = {
    "spcfw": 20000,    # e.g. 2000
    "tip3p": 20000,    # e.g. 2000
}
MSD_FIT_END_FS = {
    "spcfw": 90000,    # e.g. 30000
    "tip3p": 90000,    # e.g. 30000
}
```

Production NVE - Discussion exercise 2

➔ **Model comparison.** Which model reproduces the experimental D = 2.30 x 10^-5 cm²/s better? Is this result surprising given that TIP3P was parameterised to reproduce structural properties, not dynamical ones?

SPC/Fw. TIP3P was parameterised on structural and thermodynamic targets; its rigidity and point-charge simplicity produce artificially fast diffusion.

Rotational relaxation time

Rotational relaxation time measures how fast water molecules lose memory of their initial orientation.

➔ It is obtained from the decay of an **orientational autocorrelation function**, usually built from the molecular dipole or O–H vector.

➔ It probes **rotational dynamics** and **hydrogen-bond** network rearrangement.

$$C_2(t) = \langle P_2[\mathbf{u}(t) \cdot \mathbf{u}(0)] \rangle = \left\langle \frac{1}{2} \left(3 \left[\mathbf{u}(t) \cdot \mathbf{u}(0)\right]^2 - 1 \right) \right\rangle$$

u = molecular unit vector associated with the chosen axis
P_2 is the second Legendre polynomial function

Production NVT - Exercise 3 (~20 min)

NVT via NH and Langevin - computing different observables (g(r), vacf,msd, rotational relaxation times) with different ensembles/setups

Folder: day-07-md-simulations/exercise_3

Some questions (in the notebook as well):

- **Energy and temperature.** Compare the amplitude of T(t) fluctuations between NVE, NH, and Langevin. Which thermostat gives the tightest control? Is tight control always desirable for transport properties?
- **Structure.** Are the g(r) peaks thermostat-dependent? What does this tell you about thermostat effects on equilibrium vs. dynamical properties?
- **The effect of thermostats.** Does the choice of thermostat affect D or the rotational relaxation time? Which run should be most reliable for computing them?
- **Model comparison.** Which water model gives D and rotational relaxation time closer to experiment? Are the two models consistently better/worse, or does the answer depend on the observable?

Comparison of thermostats

When to use Nosé–Hoover

➔ **Deterministic thermostat:** controls temperature without adding stochastic noise.

➔ **Less intrusive dynamics:** often preferred over Langevin when dynamical correlations should be less perturbed.

➔ **Requires tuning**: damping time / thermostat mass must be chosen carefully, and performance can be poor for small or weakly coupled systems.

When to use Langevin

➔ **Stochastic thermostat:** applies friction and random noise to control temperature.

➔ **Robust equilibration:** useful for fast thermalization and dissipating excess energy.

➔ **Stabilizes difficult systems:** works well for small systems or poor energy redistribution.

➔ **Perturbs dynamics:** less suitable for production runs, due to artificial friction and random forces

Production NVT - Discussion exercise 3

➔ **Energy and temperature.** Compare the amplitude of T(t) fluctuations between NVE, NH, and Langevin. Which thermostat gives the tightest control? Is tight control always desirable for transport properties?

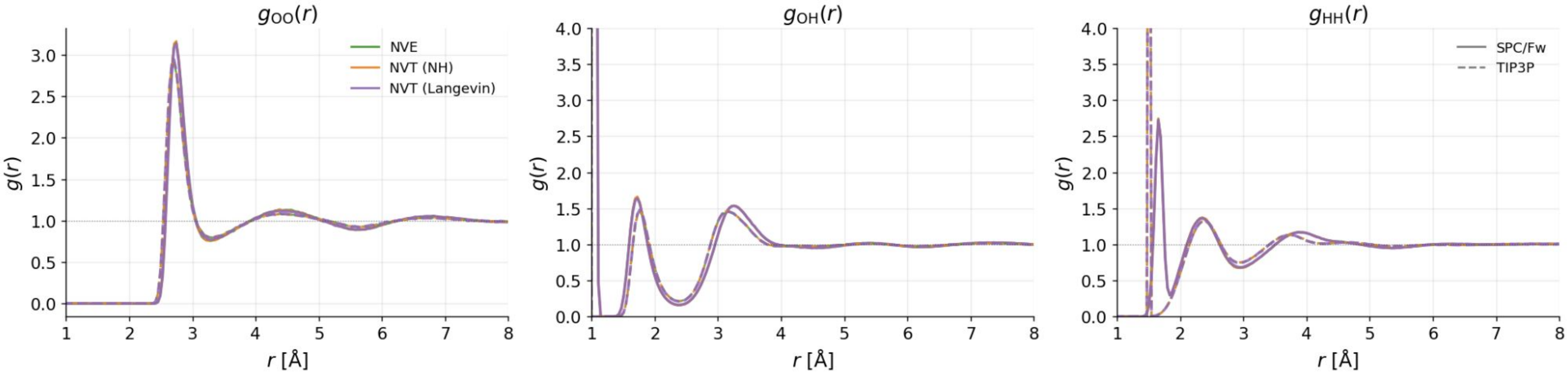
Not so evident. Langevin gives the tightest T control (friction acts on every atom locally), the friction can distort dynamical observables

```
SPC/Fw NVE: mean(T) = 297.58 K; std(T) = 7.01 K
SPC/Fw NVT (NH): mean(T) = 298.27 K; std(T) = 9.70 K
SPC/Fw NVT (Langevin): mean(T) = 299.75 K; std(T) = 9.05 K
TIP3P (rigid) NVE: mean(T) = 300.41 K; std(T) = 13.83 K
TIP3P (rigid) NVT (NH): mean(T) = 298.87 K; std(T) = 16.44 K
TIP3P (rigid) NVT (Langevin): mean(T) = 298.46 K; std(T) = 14.97 K
```

Production NVT - Discussion exercise 3

➔ **Structure.** Are the g(r) peaks thermostat-dependent? What does this tell you about thermostat effects on equilibrium vs. dynamical properties?

No, thermostats affect dynamical observables but not structural (equilibrium) ones

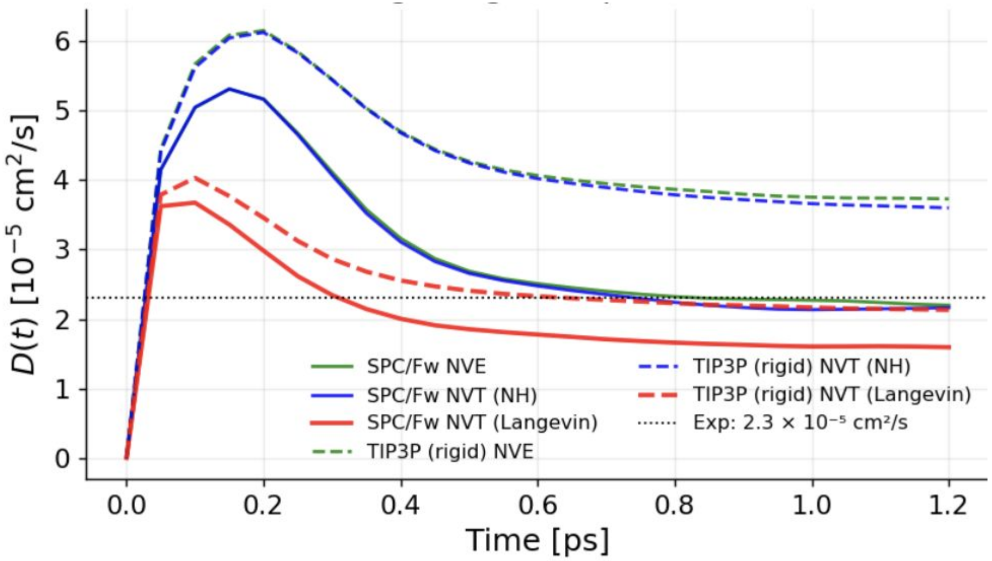


Production NVT - Discussion exercise 3

➔ **The effect of thermostats.** Does the choice of thermostat affect **D** or **rotational relaxation time**? Which run should be most reliable for computing them?

NVE is the gold standard:

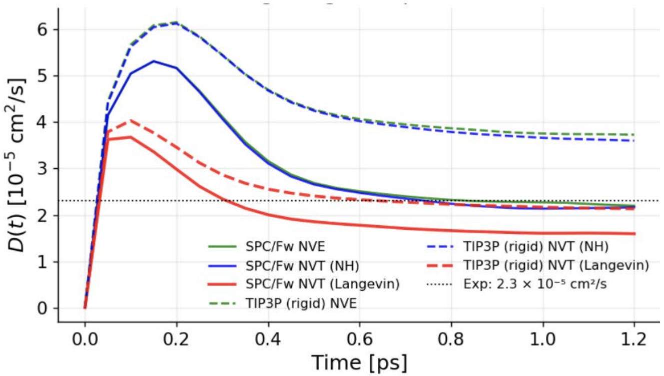
no artificial forces!

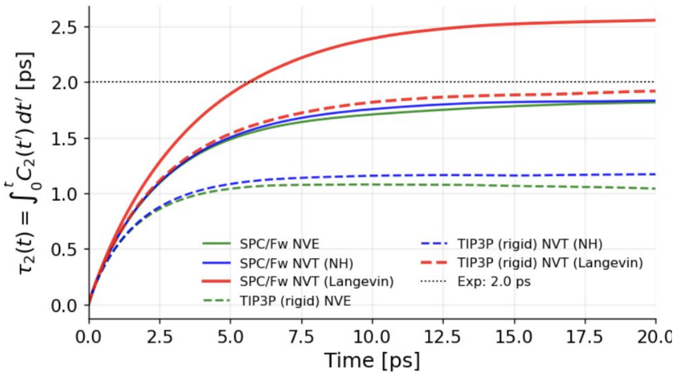


Production NVT - Discussion exercise 3

- **Model comparison.** Which water model gives the values of the diffusion coefficient & rotational relaxation times closer to experiment? Are the two models consistently better/worse, or does the answer depend on the observable?

SPC/Fw is better for both observables. The ranking is consistent across both diffusion coefficients and rotational relaxation times.





Extra: how can we estimate the error?

VACF - Block averaging

1. Divide the trajectory into N blocks.

2. Compute the VACF in each block.

3. Integrate each VACF to obtain D_i.

4. Compute:

$$\bar{D} = \frac{1}{N}\sum_{i=1}^{N} D_i$$

$$s_D = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(D_i - \bar{D})^2}$$

$$\sigma_D = \frac{s_D}{\sqrt{N}}$$